Nhill 3.0 — Configuration guide

Français : ../fr/guide-configuration.md

Three levels of settings, from infrastructure to daily use:

1. **Server** — file `/etc/nhill/server.env` (§ 1);
2. **Applications** — build options and environment variables (§ 2);
3. **Platform and businesses** — Nhill Console and Nhill Business screens (§ 3).

The marketing website has its own, separate configuration: `website/README.en.md` § 3.

1. Server: `/etc/nhill/server.env`

Secret-free template: `config/server.env.example` (installed in `/usr/share/nhill/examples/`). The file belongs to `nhill`, mode **600**: the server refuses to start if others can read it (`NH_ALLOW_INSECURE_CONFIG` lifts this check, for tests only). Every key can also be given as an environment variable. After a change: `sudo systemctl restart nhill-server`, then `sudo -u nhill nhill-server --config /etc/nhill/server.env check`.

Key	Default	Purpose
`NH_DATABASE_URL`	—	connection of the application role `nhill_app` (subject to RLS). Written by the setup wizard
`NH_MIGRATION_DATABASE_URL`	—	connection of the owner role `nhill_owner` (migrations at start-up)
`NH_DB_PROVIDER`	—	`supabase` or `postgresql`, written by setup
`NH_DB_POOL_SIZE`	= `NH_THREADS`	open connections to the database
`NH_LISTEN_ADDRESS`	`127.0.0.1`	listen address (stay on loopback behind Caddy)
`NH_LISTEN_PORT`	`8080`	listen port
`NH_THREADS`	`16`	HTTP worker threads
`NH_TRUST_PROXY`	`0`	`1` behind a trusted proxy: the client address is the **rightmost** `X-Forwarded-For` entry (2.0)
`NH_PUBLIC_BASE_URL`	—	public address `https://api.<domain>` (webhooks, payment return pages)
`NH_SECRET_KEY`	—	64 hexadecimal characters (`openssl rand -hex 32`), generated by the package. **Back it up separately**
`NH_WEBHOOK_SECRET_<PROVIDER>`	—	webhook secret (≥ 32 characters) of a generic provider

Key	Default	Purpose
`NH_ACCESS_TOKEN_MINUTES`	`15`	access token lifetime
`NH_REFRESH_TOKEN_DAYS`	`30`	refresh token lifetime (rotation, reuse detection)
`NH_ARGON2_OPS`	`3`	Argon2id cost (iterations)
`NH_ARGON2_MEM_KIB`	`65536`	Argon2id memory (64 MiB)
`NH_JOBS_INTERVAL_SECONDS`	`300`	background jobs (code rotation, subscriptions); `0` disables them
`NH_EMAIL_TRANSPORT`	`log`	`smtp`: real delivery through your relay (2.1); `log`: nothing is sent, a masked line is written to the log (development; **password reset codes are then never delivered**); `disabled`: no e-mail
`NH_SMTP_URL`	—	SMTP relay: `smtps://smtp.example.com:465` (implicit TLS) or `smtp://smtp.example.com:587` (STARTTLS **required**); never put credentials in the URL
`NH_SMTP_USERNAME`	—	relay account (optional, depending on the relay)
`NH_SMTP_PASSWORD_FILE`	—	file (mode 600) holding the relay password; otherwise `NH_SMTP_PASSWORD` in `server.env` (mode 600). Never logged
`NH_SMTP_FROM`	—	sender, for example `Nhill <no-reply@example.com>` (an address the relay accepts: SPF/DKIM of the domain)
`NH_SMTP_TLS`	`required`	`off` is accepted **only** for a local relay (`127.0.0.1`, `::1`); refused at start-up otherwise
`NH_SMTP_CA_FILE`	—	certificate authorities to use instead of the system ones (internal relay)
`NH_SUPPORT_EMAIL`	—	support address given in the signature of automatic e-mails (Nhill Solution platform: `support@nhill.online`); without it, the signature asks not to reply
`NH_SMTP_TIMEOUT_SECONDS`	`20`	relay connection timeout
`NH_PROFILE_DIR`, `NH_I18N_DIR`, `NH_FONT_DIR`, `NH_MIGRATIONS_DIR`	`/usr/share/nhill/…`	data files installed by the package
`NH_SETUP_DB_PASSWORD`	—	only for command-line `nhill-server setup`; never kept

Recommendations: do not lower `NH_ARGON2_*` below the defaults; measure sign-in time after raising them. Keep `NH_LISTEN_ADDRESS=127.0.0.1` and expose only Caddy. `NH_TRUST_PROXY=1` with no proxy in front would let a client choose the address seen by rate limits.

Supabase

Use the **Session pooler** (5432). No `anon` or `service_role` key is used. The Supabase security advisor can be checked after setup; Nhill creates a `nhill` schema not exposed by the REST API.

2. Applications

2.1 Build options (CMake)

Option	Default	Effect
`NHILL_BUILD_SERVER`	`ON` on Linux	server (Linux only)
`NHILL_BUILD_CLIENT` / `NHILL_BUILD_TESTS`	`ON`	applications / tests
`NHILL_WARNINGS_AS_ERRORS`	`OFF`	warnings as errors (CI)

No server address is compiled into the applications (3.0.0): Nhill Business asks for it when a business is created, Nhill Console and Nhill POS at their first sign-in. The address entered is checked (`GET /api/v1/meta`, HTTPS required except on `localhost`), saved on the device and can be changed later.

2.2 Environment variables (desktop)

Variable	Values	Effect
`NHILL_SERVER`	URL	server to use instead of the address saved on the device (trials, staging)
`NHILL_LANG`	`fr`, `en`	interface language at start-up
`NHILL_THEME`	`light`, `dark`, `system`	theme at start-up
`QT_SCALE_FACTOR`	e.g. `1.25`	global enlargement (standard Qt variable)

The `NHILL_TOUR_*` and `NHILL_SETUP_TOUR` variables are only used for automatic screenshots (development).

The client refuses `http://` addresses except `localhost`, `127.0.0.1`, `::1` and `10.0.2.2` (Android emulator).

2.3 User preferences

Settings › My account: language, theme. Window size and position are computed at every start from the available screen.

3. Platform and businesses

3.1 Nhill Console (creators)

Screen	Settings
Platform settings	trial length (30 days by default), grace period, reminder, establishment code rotation (30 days), limits, payment instructions; 2.1: `password_reset_minutes` (validity of the e-mailed code, 20 min, 5 to 60), `password_reset_max_attempts` (wrong codes before the code is cancelled, 5, 1 to 10), `temporary_password_hours` (validity of a temporary password, 72 h, 1 to 168)

Screen	Settings
Plans & prices	plans: 9 features that can be switched on and 6 limits (users, tills, warehouses, products, documents per month, storage), terms (trial, grace, reminders, self-service, automatic renewal, fallback plan), **prices per duration and currency** — nhill-console.md § 5
Assignment rules	plan given automatically to a new business (type, country, trial or granted period) — nhill-console.md § 6
Promo codes	launch or partnership discounts
Payment providers	credentials (encrypted, never shown again), test/production mode, webhook address

3.2 Nhill Business (business)

Tab	Main settings
Business	identity, time zone, legal identifiers, logo, document language, payment terms, tax rounding, payment instructions, notices printed on every document, footer, negative stock, full data export
Taxes and accounts	tax rates (16 % VAT in the DRC profile; a rate cannot be edited), cash and bank accounts, dated exchange rates
Point of sale	tills, authorised cashiers, payment methods, maximum discount without approval, 58/80 mm receipt
Team	roles, invitations, resetting a member's password

3.3 Country profiles and templates

`server/profiles/*.json`: tax profiles (rates, mentions, numbering) synchronised into the database at start-up. An issued document keeps the version of the profile applied. Details: `docs/archive/modeles-et-profils.md` (French). The tax values of the profiles must be checked by a professional before use (see legal-parameters.md).

4. Password rule (2.0, unchanged in 2.1)

Item	Value	Where
Minimum length	**8 characters** (Unicode: an "é" counts as one)	`common/include/nhill/validation.hpp` → `kMinPasswordLength`
Maximum length	128 bytes	`kMaxPasswordBytes`
Refused	common passwords (built-in list), same as the e-mail, repetitive	`common/src/validation.cpp`
Periodic expiry	**none**	—
Storage	Argon2id (libsodium), never in plain text	server
Two-factor authentication	mandatory for Nhill Console; not imposed on businesses	—
Temporary password (2.1)	12 characters drawn from an unambiguous alphabet (`xxxx-xxxx-xxxx`), shown once to the administrator, valid `temporary_password_hours`; the next sign-in requires a new password, which must differ from it	`server/src/services/members.cpp`

The same function is called by the applications (before sending) and by the server (which decides). Changing the rule requires rebuilding the server **and** the applications.

5. Forgotten password and remembered session (2.1)

Item	Behaviour	Setting
Employee of an establishment	the administrator gives a temporary password (Settings › Team); change required at the next sign-in; the member's sessions are closed	`temporary_password_hours`
Owner, administrator, account of several establishments	8-digit one-time code by e-mail, same answer whatever the address	`password_reset_minutes`, `password_reset_max_attempts`
Rate limits	code requests: 3 per hour per address, 10 per hour per IP address; checks: 10 per 15 min per address, 20 per IP; resets by an administrator: 10 per 15 min	fixed (server)
E-mails	code, "contact your administrator" instructions, notices of reset and of password change, in French or English according to the account	`NH_EMAIL_TRANSPORT=smtp` and `NH_SMTP_*`
"Remember me"	refresh token kept by QtKeychain: Windows Credential Manager, Android Keystore, Secret Service/KWallet on Linux; never a plain-text fallback; the yes/no preference is kept in the application settings	tick box at sign-in, Settings › My account

Enabling mail: without `NH_EMAIL_TRANSPORT=smtp`, owners and administrators cannot recover a forgotten password by themselves (the server says so at start-up). Check the configuration with a test account: "Forgot your password?" must deliver a code within seconds; on failure, the relay's error (without the password) appears in the server log and in `email_outbox.last_error`.